

Assignment Title: Fuzzy Logic–Based Room Temperature Control

Objective: To design a fuzzy inference system (FIS) that controls a fan or air conditioner speed based on **room temperature** and **humidity**.

Problem Statement: You are tasked to develop a fuzzy logic system that adjusts fan speed in a room depending on two input variables:

1. **Temperature** (Low, Medium, High)
2. **Humidity** (Dry, Comfortable, Humid)

The **output** is:

- **Fan Speed** (Slow, Medium, Fast).

Tasks for Students:

1. **Define linguistic variables and membership functions** for temperature, humidity, and fan speed.
 - e.g., Temperature: Low (0–15°C), Medium (15–30°C), High (30–45°C).
 - e.g., Humidity: Dry (0–30%), Comfortable (30–60%), Humid (60–100%).
 - Fan Speed: Slow (0–40), Medium (40–70), Fast (70–100).
 2. **Construct fuzzy rules** (e.g., If temperature is high AND humidity is humid THEN fan speed is fast).
 3. **Implement the fuzzy system** using any tool (C/C++, Python with scikit-fuzzy, or even manual calculation).
 4. **Test the system** with at least **5 input cases** and record the output fan speed.
 5. **Write a short report** (2–3 pages) describing membership functions, rule base, and test results.
-

Khulna University of Engineering & Technology (KUET)
Department of Computer Science and Engineering
CSE 4109: Artificial Intelligence (Topic: Fuzzy Expert Systems)
Instructor: Prof Dr. M.M.A. Hashem

Assignment 1: Fuzzy Room Temperature Control

Objective:

Design a fuzzy system to control fan speed based on room temperature and humidity.

Inputs & Outputs:

- Inputs:
 - Temperature (Low, Medium, High)
 - Humidity (Dry, Comfortable, Humid)
- Output:
 - Fan Speed (Slow, Medium, Fast)

Tasks:

1. Define membership functions for temperature, humidity, and fan speed.
2. Construct at least **6 fuzzy rules** (e.g., If temperature is high AND humidity is humid THEN fan speed is fast).
3. Implement the system in C/C++/MATLAB/Python or manually.
4. Test with **5 input cases**.
5. Submit a **short report (2–3 pages)**.

Khulna University of Engineering & Technology (KUET)
Department of Computer Science and Engineering
CSE 4109: Artificial Intelligence (Topic: Fuzzy Expert Systems)
Instructor: Prof Dr. M.M.A. Hashem

Assignment 2: Fuzzy Traffic Light Control

Objective:

Control traffic light timing dynamically using fuzzy logic.

Inputs & Outputs:

- Inputs:
 - Vehicle Density (Low, Medium, High)
 - Waiting Time (Short, Medium, Long)
- Output:
 - Green Light Duration (Short, Medium, Long)

Tasks:

1. Define fuzzy sets for inputs and output.
2. Develop **at least 6 rules** (e.g., If density is high AND waiting time is long THEN duration is long).
3. Simulate different traffic conditions.
4. Document results.

Khulna University of Engineering & Technology (KUET)
Department of Computer Science and Engineering
CSE 4109: Artificial Intelligence (Topic: Fuzzy Expert Systems)
Instructor: Prof Dr. M.M.A. Hashem

Assignment 3: Fuzzy Student Performance Evaluation

Objective:

Design a fuzzy evaluation system to assess student performance.

Inputs & Outputs:

- Inputs:
 - Exam Score (Poor, Average, Good)
 - Class Participation (Low, Medium, High)
 - Assignment Quality (Low, Medium, High)
- Output:
 - Performance (Weak, Moderate, Strong)

Tasks:

1. Define fuzzy sets for inputs and output.
2. Create **at least 9 rules** (covering all major cases).
3. Test the system on **5 sample students**.
4. Write a **summary report**.

Khulna University of Engineering & Technology (KUET)
Department of Computer Science and Engineering
CSE 4109: Artificial Intelligence (Topic: Fuzzy Expert Systems)
Instructor: Prof Dr. M.M.A. Hashem

Assignment 4: Fuzzy Washing Machine Controller

Objective:

Design a fuzzy system that controls wash time based on clothes condition.

Inputs & Outputs:

- Inputs:
 - Dirtiness (Low, Medium, High)
 - Load Size (Small, Medium, Large)
- Output:
 - Wash Time (Short, Medium, Long)

Tasks:

1. Define fuzzy sets for dirtiness, load size, and wash time.
2. Create **at least 6 fuzzy rules**.
3. Simulate and test with **5 cases**.
4. Submit findings in a report.